

Solution:

$$C_D = \frac{2P}{\rho A v^3} = \frac{2(294)}{1000(0.20)(1.4)^3}$$

$$C_D = 1.07$$

$$\frac{\Delta C_D}{C_D} = \frac{\Delta P}{P} + \frac{\Delta \rho}{\rho} + \frac{\Delta A}{A} + 3 \frac{\Delta v}{v} = \frac{2}{294} + \frac{1}{1000} + \frac{0.01}{0.20} + 3 \frac{0.1}{1.4}$$

$$\frac{\Delta C_D}{C_D} = 0.272$$

$$\Delta C_D = 0.272 \times 1.07 \approx 0.3$$

$$C_D = 1.1 \pm 0.3$$

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- (c) (i) Derive, from the definition of power, an expression of the drag force F experienced by the swimmer in terms of the velocity v .

[2]

Solution:

Since $P = \frac{W}{t} = \frac{Fs}{t} = Fv$

$$F = \frac{P}{v} = \frac{\frac{1}{2} C_D \rho A v^3}{v}$$

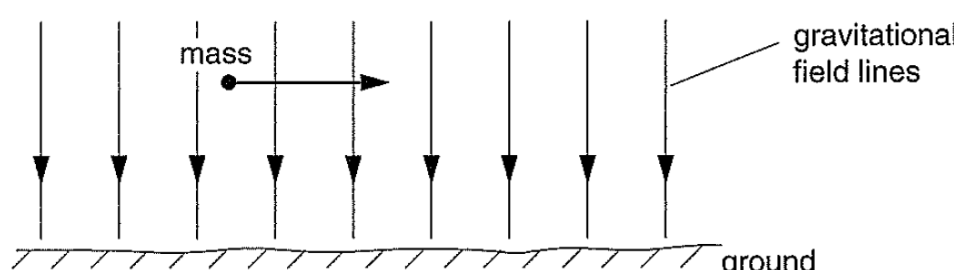
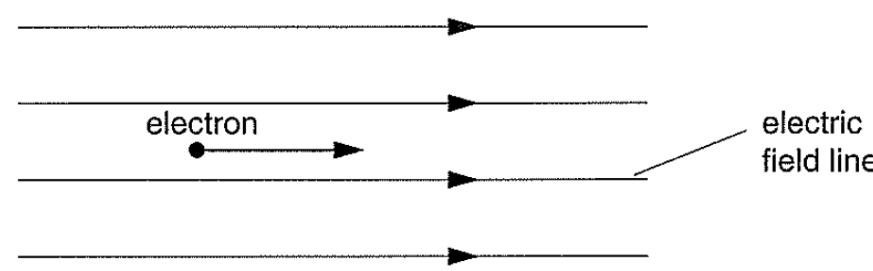
$$F = \frac{1}{2} C_D \rho A v^2$$

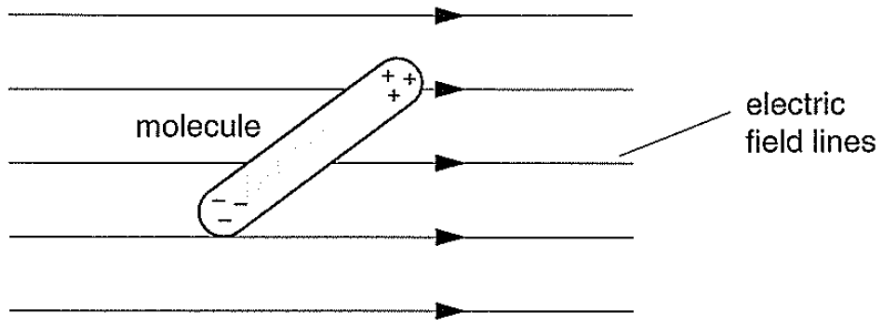
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- (ii) Hence or otherwise, calculate the work done by the swimmer to overcome the drag force when he swims a distance of 50 m at a constant speed of 1.4 m s⁻¹.

				work done = J [2]
			Solution: $W = Fs = \frac{1}{2} C_D \rho A v^2 s = \frac{1}{2} (1.07)(1000)(0.20)(1.4)^2 (50)$ $W = 10500 \text{ J}$	1 1

2	(a)	Fig. 2.1 shows a mass initially travelling at right angles to the Earth's uniform gravitational field.  Fig. 2.1 Describe the subsequent motion of the mass.	
		 [1]	
		Solution: Mass will follow a <u>parabolic</u> path towards the ground	1
	(b)	Fig. 2.2 shows an electron initially travelling parallel to a uniform electric field.  Fig. 2.2 Describe the subsequent motion of the electron.	

		[2]	
		Solution: Electron will slow down to a stop and then accelerate to the left	1 1
	(c)	Fig. 2.3 shows a long molecule placed in a uniform electric field.  Fig. 2.3 The ends of the molecule have equal but opposite charges. Describe and explain the initial motion of the molecule in the electric field.	
		 [2]	
		Solution: Forces acting on ends of the molecule are equal and opposite or they form a couple This causes the molecule to rotate clockwise about its centre of mass	1 1
	(d)	Fig. 2.4 shows a sphere of weight 1.6×10^{-2} N with an electric charge of $+2.0 \mu\text{C}$. It is released from rest, in vacuum, between two parallel, vertical metal plates. The separation of the plates is 0.10 m. One plate has a potential of +80 V and the other plate has a potential of -80 V.	

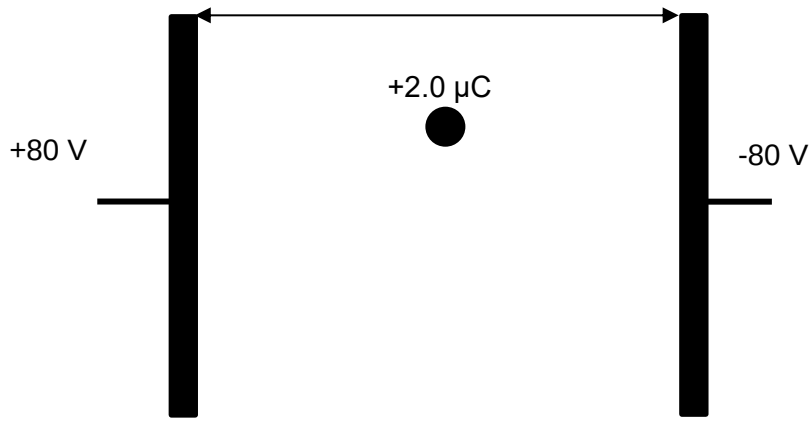


Fig. 2.4

- (i) Determine the electric force experienced by the sphere.

force = N [2]

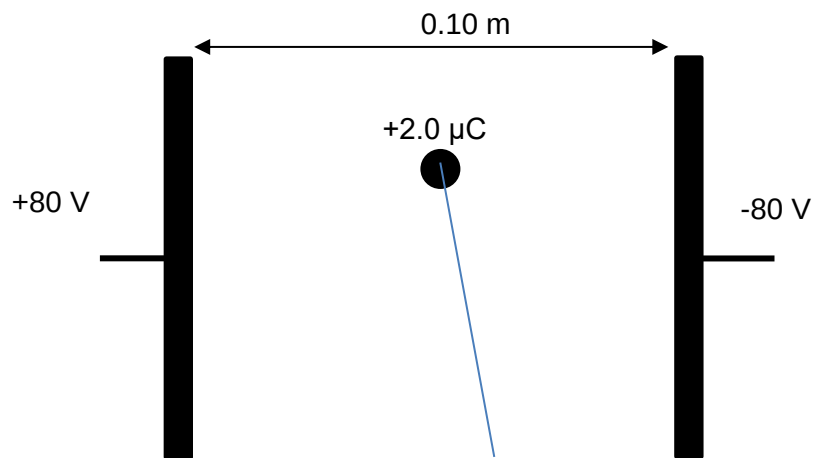
Solution:

$$\begin{aligned} F_E &= qV/d \\ &= 2 \times 10^{-6} \times 160 / 0.10 \\ &= 3.2 \times 10^{-3} \text{ N} \end{aligned}$$

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- (ii) On Fig. 2.4, sketch the path taken by the sphere after it is released. [1]

Solution:



Straight line pointing downward slightly to right (less than 20 degrees from vertical) 1

- (e) The variations with separation of the gravitational potential energy U_G and of the electric potential energy U_E between two protons are shown in Fig. 2.5.

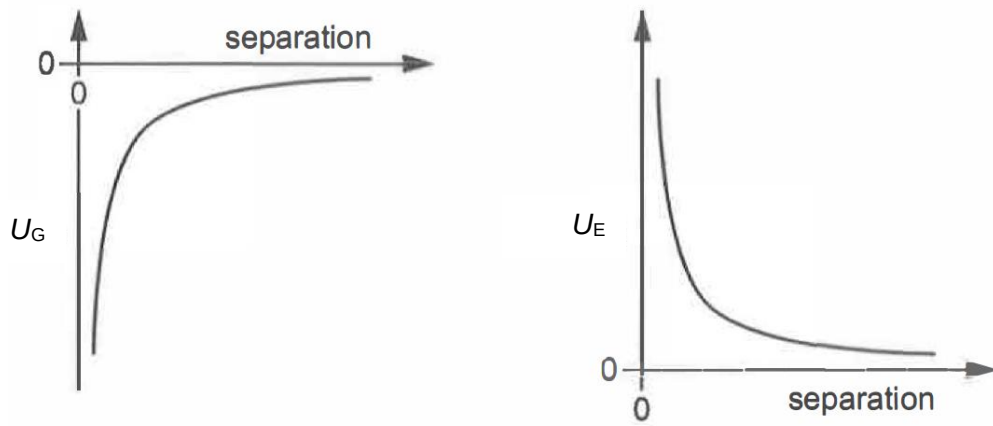


Fig. 2.5

Explain why the gravitational potential energy and the electric potential energy have opposite signs.

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..... [2]

Solution:

The gravitational force acting between the protons are attractive whereas the electric force acting between the protons are repulsive,

1

so

the work done by an external agent to bring a proton from infinity to the point is negative.

1

the work done by an external agent to bring a proton from infinity to the point is positive.

3 A cantilever open cable stayed bridge is a unique yet functional variation of the traditional cable